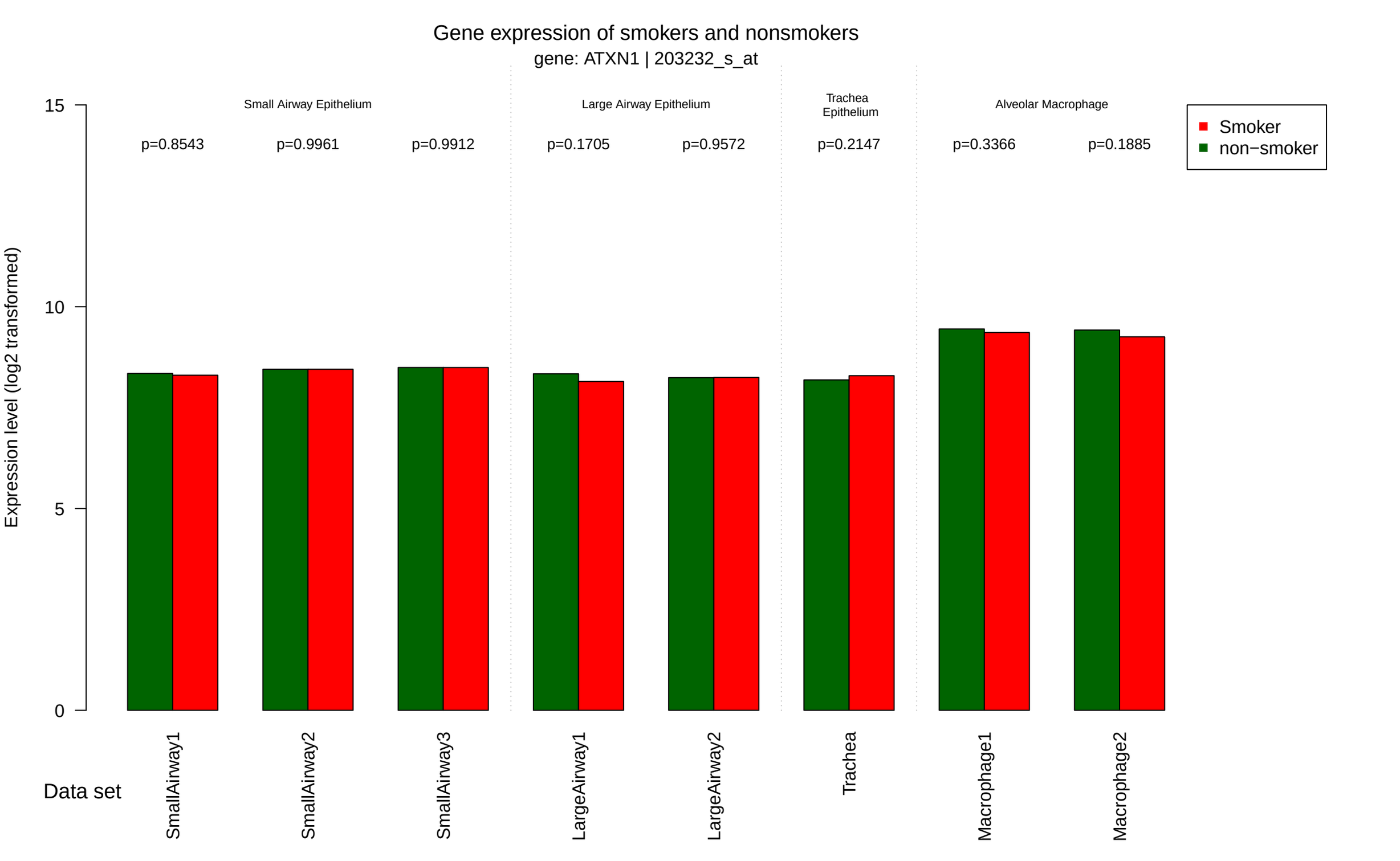
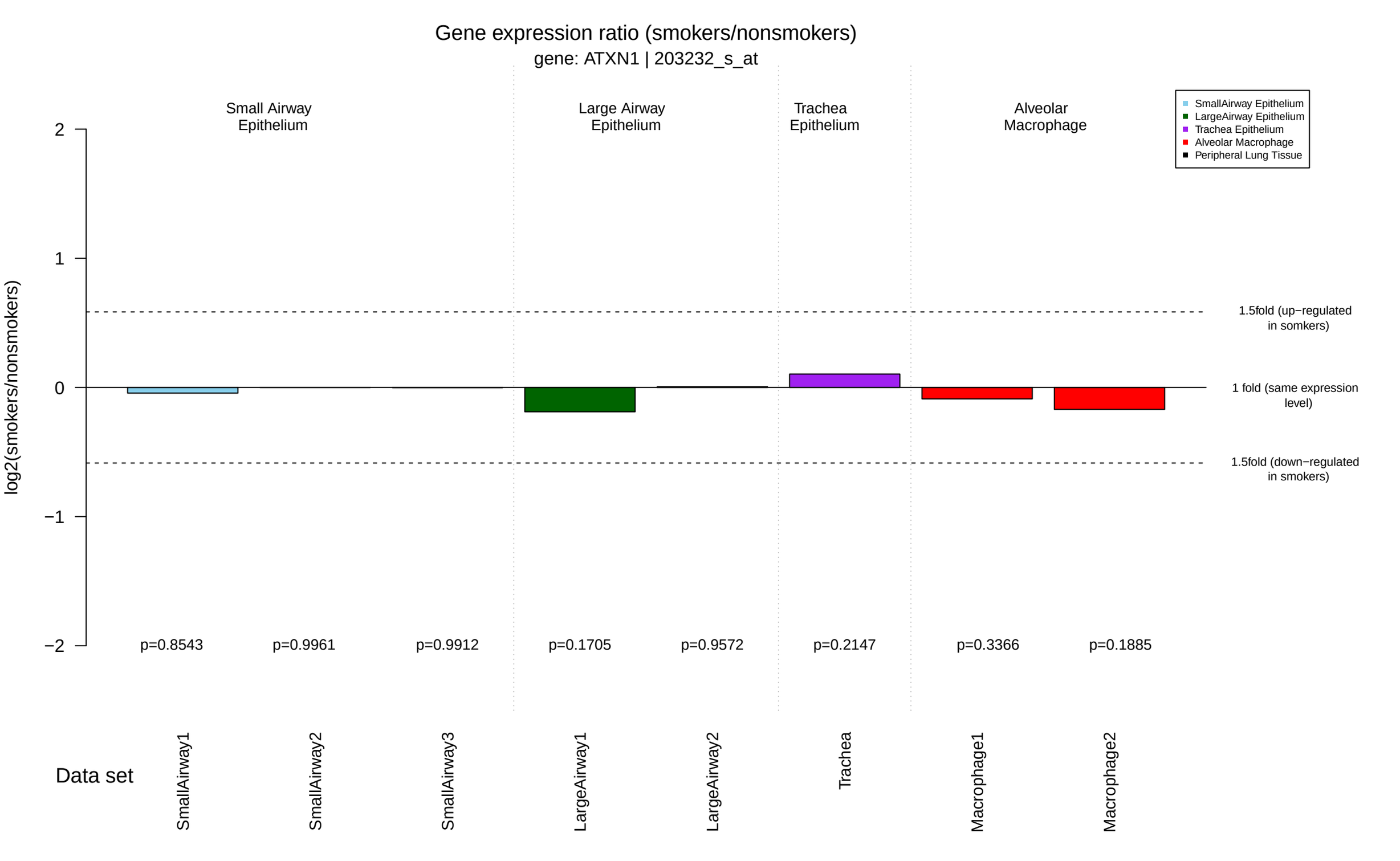
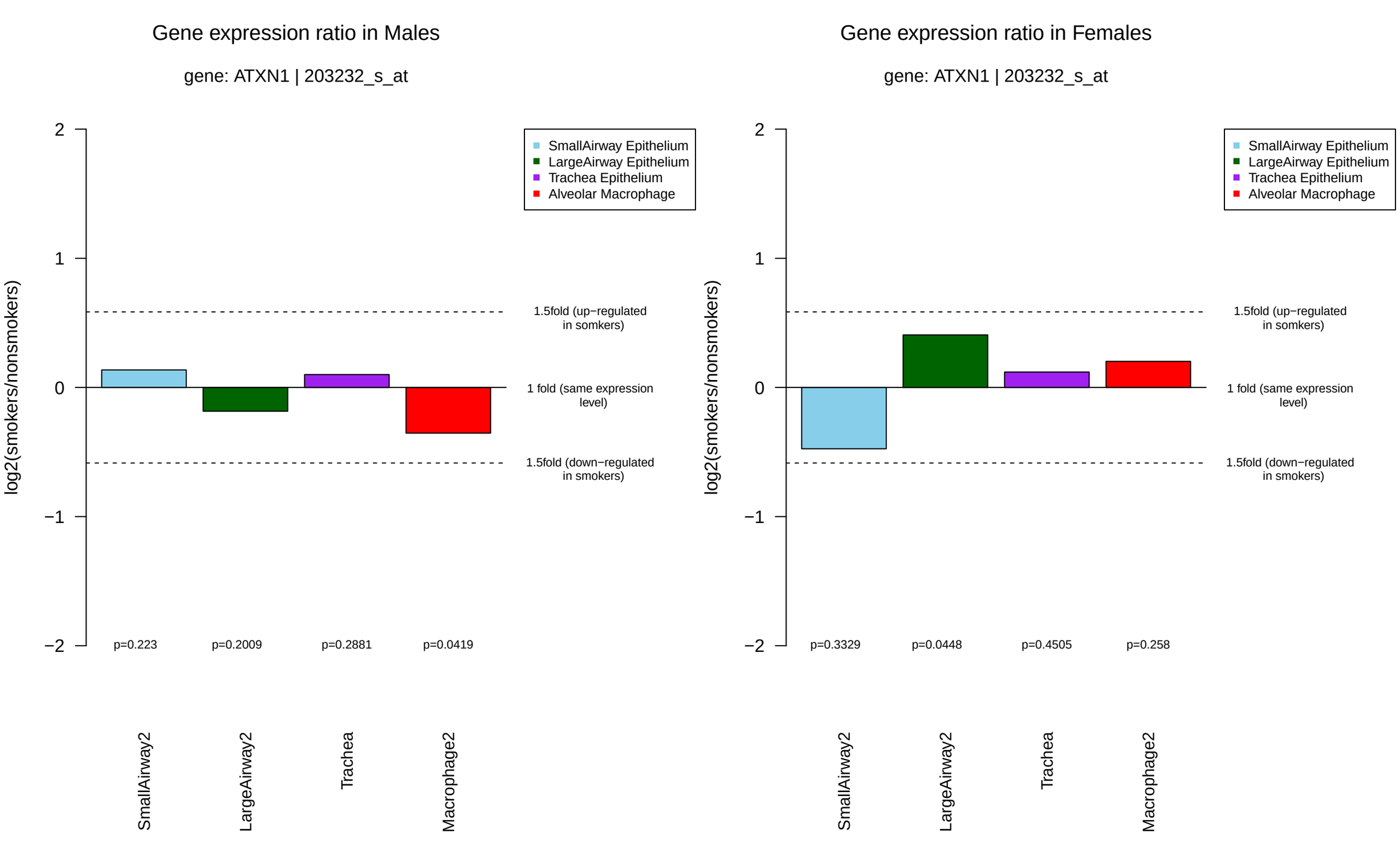
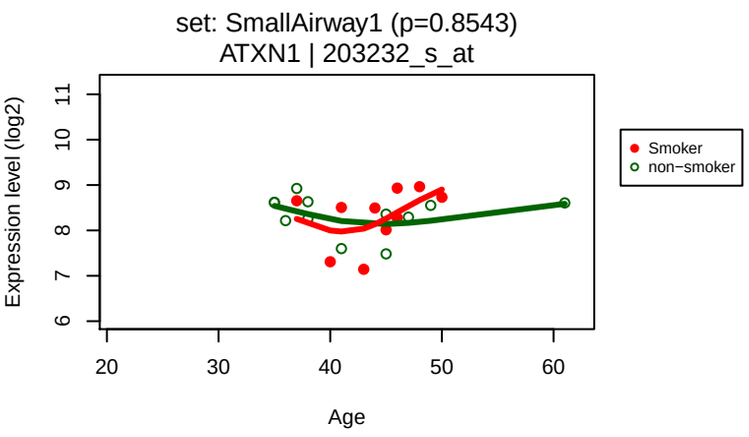
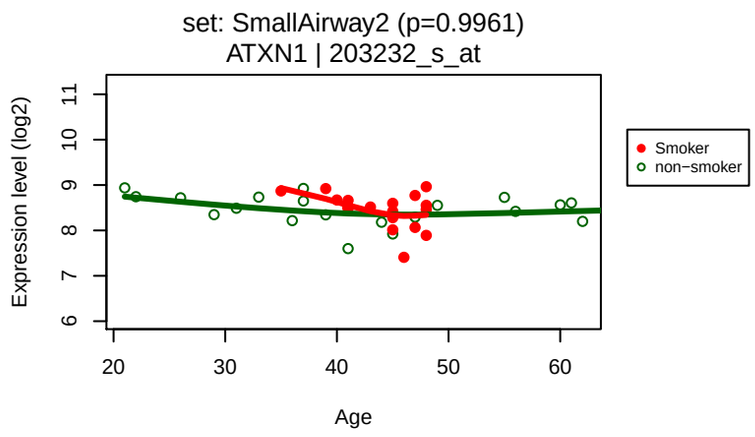
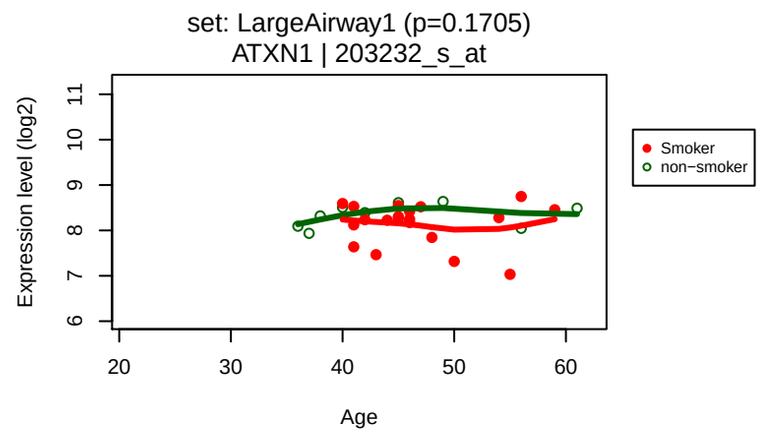

Gene expression of smokers and nonsmokers
gene: ATXN1 | 203232_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8543
p=0.9961
p=0.9912
p=0.1705
p=0.9572
p=0.2147
p=0.3366
p=0.1885
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

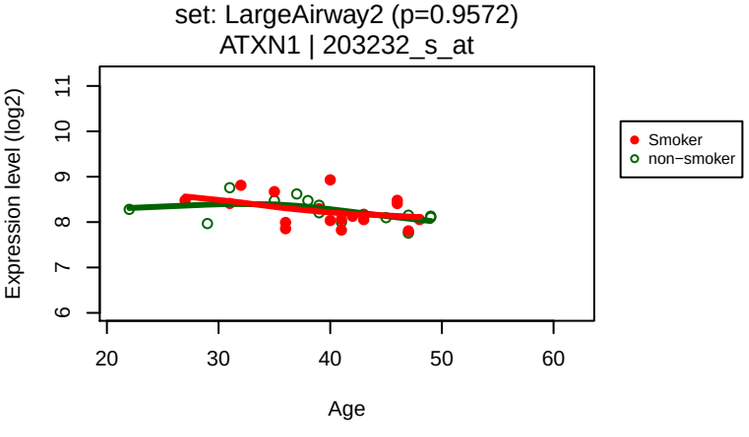

Gene expression ratio (smokers/nonsmokers)
gene: ATXN1 | 203232_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8543
p=0.9961
p=0.9912
p=0.1705
p=0.9572
p=0.2147
p=0.3366
p=0.1885
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

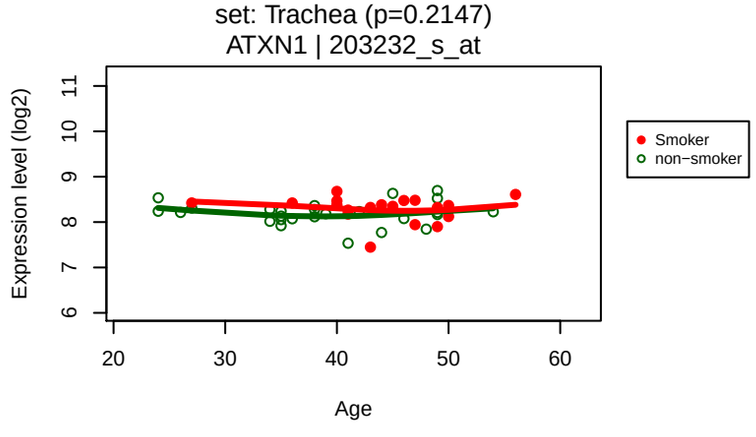

Gene expression ratio in Males
gene: ATXN1 | 203232_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.223
p=0.2009
p=0.2881
p=0.0419
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATXN1 | 203232_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3329
p=0.0448
p=0.4505
p=0.258
SmallAirway2
LargeAirway2
Trachea
Macrophage2

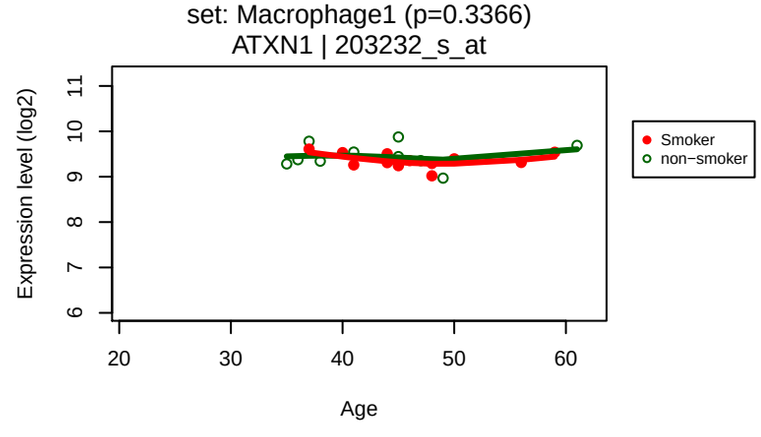

set: SmallAirway1 (p=0.8543)
ATXN1 | 203232_s_at
Expression level (log2)
Age
Smoker
non−smoker

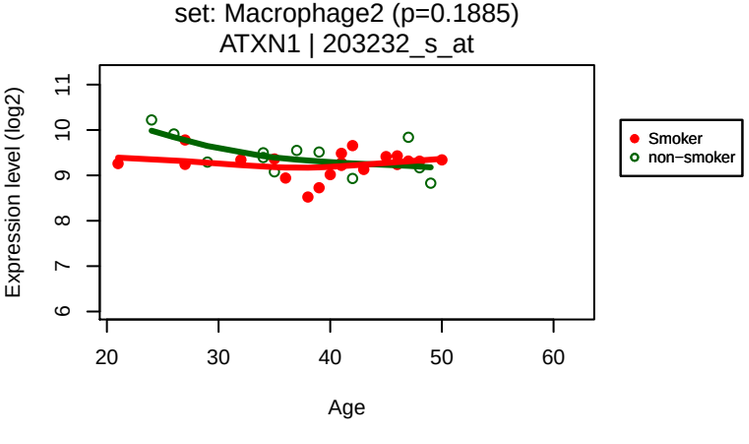

set: SmallAirway2 (p=0.9961)
ATXN1 | 203232_s_at
Expression level (log2)
Age
Smoker
non−smoker

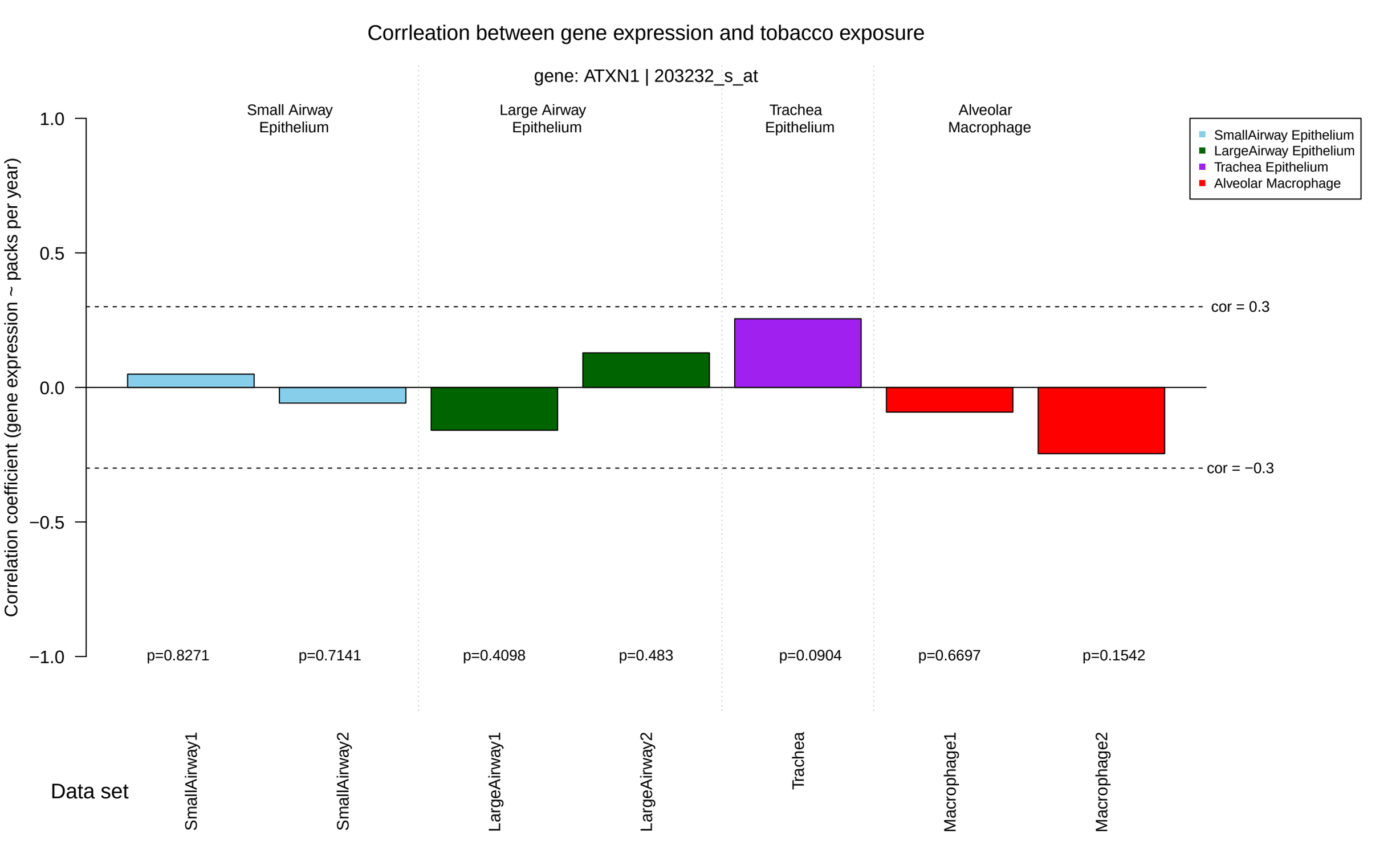

set: LargeAirway1 (p=0.1705)
ATXN1 | 203232_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9572)
ATXN1 | 203232_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2147)
ATXN1 | 203232_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3366)
ATXN1 | 203232_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1885)
ATXN1 | 203232_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATXN1 | 203232_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8271
p=0.7141
p=0.4098
p=0.483
p=0.0904
p=0.6697
p=0.1542
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2